Explicit infinity point: checked proof boundary

aburan28/crypto, genus-2 projective-law follow-up, 25 October 2026

Object. For a field K and arbitrary $a_3, a_2, a_1, a_0 \in K$, the explicit infinity chart has ring

$$R = K[z, w]/\big(w^2 - z(1 + a_3 z^2 + a_2 z^3 + a_1 z^4 + a_0 z^5)\big).$$

The checked overlap with the affine chart is the principal open $D(z)$.

Checked result. The complement of $D(z)$ in $\operatorname{Spec} R$ contains exactly one point, the origin ideal (z, w). For any prime $\mathfrak{p}$ containing z, the equation gives $w^2 \in \mathfrak{p}$, and primality gives $w \in \mathfrak{p}$. The pre-existing Lean theorem proves (z, w) is maximal, so $\mathfrak{p} = (z, w)$. Evaluation at $z = w = 0$ proves the origin is outside $D(z)$. This statement is for the explicit affine chart; it is not a claim of projectivity or smoothness of the glued scheme.

Compiler evidence. Pinned Lean v4.34.1 and Mathlib revision

`d13f23b723b8a846827a245b89c10fc7d3f11612`

The command `timeout 180s lake env lean Common22CompletionBridge.lean` exited 0. The four `#print axioms` lines list exactly `propext`, `Classical.choice`, and `Quot.sound`. The adjacent CSV records each theorem, its hypotheses, and status; the SVG shows the dependency and open-obligation graph.

Chosen completion target. Let A be the graded quotient

$$K[X, Y, Z]/(Y^2 - X^5 Z - a_3 X^3 Z^3 - a_2 X^2 Z^4 - a_1 X Z^5 - a_0 Z^6), \quad (\deg X, \deg Y, \deg Z) = (1, 3, 1).$$

On $D_+(Z)$ set $x = X/Z,\ y = Y/Z^3$; on $D_+(X)$ set $z = Z/X,\ w = Y/X^3$. Their equations are the two checked affine chart equations. The equation forces $Y^2 = 0$ when $X = Z = 0$, so these two opens cover the relevant homogeneous primes. The graded construction, chart-ring isomorphisms, and overlap comparison remain unproved in Lean. The pinned Mathlib supplies Proj properness for finite-type graded rings over their degree-zero part; smoothness still needs the squarefree hypothesis and chartwise proof. The principal divisor of the interpolating function, including its order at this unique point, and the resulting Pic^0 equation remain separate proof obligations.

Interpretation. This packet advances the local geometry of the constructed chart. It does not revise measured cost ratios or assert correctness of the full published law.